ome > syeda-areej-e-fatima > Sem 2 > OOPS_labtask > lab_tasks > P25-0014_aree

```cpp
#include<iostream>
using namespace std;


int main()
{

    int campus, dep;
    cout<<"Enter number of campuses : ";
    cin>>campus;
    cout<<"Enter number of departments : ";
    cin>>dep;

    int **data = new int *[campus];

    for (int i = 0; i < campus; i++)
    {
        *(data+i)= new int[dep];
    }

```

ome > syeda-areej-e-fatima > Sem 2 > OOPS_labtask > lab_tasks > P25-0014_areej_AI-2A_Q1.cpp > main()

```cpp


    for (int i = 0; i < campus; i++)
    {
        for (int j = 0; j < dep; j++)
        {
            if(i==0)
            {
               cout<<"Enter students for Peshawar Campus " ;
                if(j==0)
                {
                    cout<<"- Computer Science : ";
                    cin>>*(*(data+i)+j);
                }
                else if(j==1)
                {
                    cout<<"- Software Engineering : ";
                    cin>>*(*(data+i)+j);

                }
                else
                {
                    cout<<"- Data Science";
                    cin>>*(*(data+i)+j);

                }


            }
            else if(i==1)
            {
                cout<<"Enter students for Islamabad Campus " ;
                if(j==0)
                {
                    cout<<"- Computer Science : ";
                    cin>>*(*(data+i)+j);
                }
```

ome > syeda-areej-e-fatima > Sem 2 > OOPS_labtask > lab_tasks > P25-0014_areej_AI-2A_Q1.cpp > main()

```
            }
            else if(i==1)
            {
                cout<<"Enter students for Islamabad Campus " ;
                if(j==0)
                {
                    cout<<"- Computer Science : ";
                    cin>>*(*(data+i)+j);
                }
                else if(j==1)
                {
                    cout<<"- Software Engineering : ";
                    cin>>*(*(data+i)+j);
                }
                else if(j==2)
                {
                    cout<<"- Data Science : ";
                    cin>>*(*(data+i)+j);
                }
                else
                {
                    cout<<"Wrong Department"<<endl;
                }

            }
            else if(i==2)
            {
                cout<<"Enter students for Lahore Campus " ;
                if(j==0)
                {
                    cout<<"- Computer Science : ";
                    cin>>*(*(data+i)+j);
                }
                else if(j==1)
                {
                    cout<<"- Software Engineering : ";
```

ome > syeda-areej-e-fatima > Sem 2 > OOPS_labtask > lab_tasks > P25-0014_areej_AI-2A_Q1.cpp > main()

```cpp
            }
            else if(i==2)
            {
                cout<<"Enter students for Lahore Campus " ;
                if(j==0)
                {
                    cout<<"- Computer Science : ";
                    cin>>*(*(data+i)+j);
                }
                else if(j==1)
                {
                    cout<<"- Software Engineering : ";
                    cin>>*(*(data+i)+j);
                }
                else if(j==2)
                {
                    cout<<"- Data Science : ";
                    cin>>*(*(data+i)+j);
                }
                else
                {
                    cout<<"Wrong Department";
                }
            }
            else
            {
                cout<<"Wrong campus";
            }


        }
        cout<<endl;

    }

```

ome > syeda-areej-e-fatima > Sem 2 > OOPS_labtask > lab_tasks > P25-0014_areej_AI-2A_Q1.cpp > main()

```cpp
    cout<<"Campuses / Departments | \t CS \t SE \t DS "<<endl;
    cout<<"-----------------------------------------------"<<endl;

    for (int i = 0; i < campus; i++)
    {
        if(i==0)
        {
            cout<<"Peshawar Campus \t\t";
        }
        else if(i==1)
        {
            cout<<"Islamabad Campus \t\t";
        }
        else if(i==2)
        {
            cout<<"Lahore Campus \t\t\t";
        }
        else
        {
            cout<<"Wrong campus"<<endl;
        }

        for (int j = 0; j < dep; j++)
        {
            if (j==0)
            {
                cout<<*(*(data+i)+j)<<"\t";
            }
            else if (j==1)
            {
                cout<<*(*(data+i)+j)<<"\t";
            }
            else if (j==2)
            {
                cout<<*(*(data+i)+j)<<"\t";
            }
```

ome > syeda-areej-e-fatima > Sem 2 > OOPS_labtask > lab_tasks > P25-0014_areej_AI-2A_Q1.cpp > main()

```
                cout<<*(*(data+i)+j)<<"\t";
            }
            else if (j==1)
            {
                cout<<*(*(data+i)+j)<<"\t";
            }
            else if (j==2)
            {
                cout<<*(*(data+i)+j)<<"\t";
            }
            else
            {
                cout<<"Wrong department"<<endl;
            }
        }

        cout<<endl;
    }



    for (int i = 0; i < campus; i++)
    {
        delete [] (data+i);
    }

    delete [] data;




    return 0;

}
```

```
Enter number of campuses : 3
Enter number of departments : 3
Enter students for Peshawar Campus - Computer Science : 120
Enter students for Peshawar Campus - Software Engineering : 95
Enter students for Peshawar Campus - Data Science60

Enter students for Islamabad Campus - Computer Science : 67
Enter students for Islamabad Campus - Software Engineering : 246
Enter students for Islamabad Campus - Data Science : 678

Enter students for Lahore Campus - Computer Science : 45
Enter students for Lahore Campus - Software Engineering : 569
Enter students for Lahore Campus - Data Science : 456

Campuses / Departments |         CS      SE      DS
-----------------------------------------------
Peshawar Campus                 120     95      60
Islamabad Campus                67      246     678
Lahore Campus                   45      569     456
munmap_chunk(): invalid pointer
Aborted (core dumped)
syeda-areej-e-fatima@velanora:~/Sem 2/OOPS_labtask/lab_tasks$
```

signment > P250014_areej_AI-2A_Q2.cpp > main()

```cpp
#include<iostream>
using namespace std;


int main()
{

    int row;
    cout<<"Enter the total number of rows";
    cin>>row;
    bool **game = new bool *[row];
    int col;
    cout<<"Enter the total number of columns";
    cin>>col;

    for (int i = 0; i < row; i++)
    {
        *(game+i) = new bool[col];
    }


    cout<<"Enter :"<<endl;
    for (int i = 0; i < row; i++)
    {
        cout<<"Enter "<<col<<" digits for row "<<i+1<<endl;

        for (int j = 0; j < col; j++)
        {
            cin>>*(*(game+i)+j);
        }
    }
```

Assignment > P250014_areej_AI-2A_Q2.cpp > main()

```cpp

    // count loop
    int count_live =0;
    for (int i = 0; i < row; i++)
    {
        for (int j = 0; j < col; j++)
        {
            if(*(*(game+i)+j) != 0)
            {
                count_live++;
            }
        }
    }
    int hit = 0;
    cout<<"total life enemies"<<count_live<<endl;

    for(int i = 0 ; hit != count_live; i++)
    {
        int row_index , col_index;
        cout<<"Enter the index of row : ";
        cin>>row_index;
        cout<<"Enter the index of colum : ";
        cin>>col_index;


        if( *(*(game+row_index-1) + col_index-1) != 0 )
        {
            *(*(game+row_index-1) + col_index-1) = 0;
            cout<<"Hit. Enemy destroyed"<<endl;
            hit++;
        }
        else
        {
            cout<<"Already dead "<<endl;
        }

    }

```

```
    cout<<"All enemies are destroyed "<<endl;
    cout<<"Total hits required: "<<hit<<endl;

    cout<<"Final grid"<<endl;

    for (int i = 0; i < row; i++)
    {
        for (int j = 0; j < col; j++)
        {
            cout<<*(*(game+i)+j);
            cout<<" ";
        }
        cout<<endl;
    }

    cout<<endl;

    for (int i = 0; i < row; i++)
    {
        delete [] (game+i);
    }

    delete [] game;



    return 0;
}
```

```
syeda-areej-e-fatima@velanora:~/Sem 2/OOPS$ cd "/home/syeda-areej-e-fatima/Sem 2/OOP
P250014_areej_AI-2A_Q2 && "/home/syeda-areej-e-fatima/Sem 2/OOPS/Assignment/"P250014
Enter the total number of rows3
Enter the total number of columns3
Enter :
Enter 3 digits for row 1
1
0
1
Enter 3 digits for row 2
1
1
0
Enter 3 digits for row 3
0
1
1
total life enemies6
Enter the index of row : 1
Enter the index of colum : 1
Hit. Enemy destroyed
Enter the index of row : 1
Enter the index of colum : 3
Hit. Enemy destroyed
Enter the index of row : 2
Enter the index of colum : 1
Hit. Enemy destroyed
Enter the index of row : 2
Enter the index of colum : 2
Hit. Enemy destroyed
Enter the index of row : 3
Enter the index of colum : 2
Hit. Enemy destroyed
Enter the index of row : 3
Enter the index of colum : 3
Hit. Enemy destroyed
All enemies are destroyed
Total hits required: 6
Final grid
0 0 0
0 0 0
0 0 0
```